Aditya Raut

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Professional Summary

Software Engineer with **3+ years of experience** building **backend services and distributed systems** in Go, Node.js/TypeScript, Python, and AWS. Led a Node.js-to-Go migration of a real-time monitoring system for GE (**40% lower memory, 3x concurrent connections**) and built fault-tolerant pipelines processing millions of records. M.S. in Computer Science; also built **multi-agent and LLM-routing tools** in Python.

Technical Skills

Languages: Python, Java, Go, TypeScript, JavaScript, SQL, C/C++
Backend & APIs: Node.js, Express.js, Spring Boot, FastAPI, Gin, Django, REST APIs, Microservices, React, JUnit
Distributed Systems: Event-Driven Architecture, Apache Kafka, RabbitMQ, Flink, Redis, WebSockets
Databases & Caching: PostgreSQL, MySQL, MongoDB, DynamoDB, Redis
Cloud & DevOps: AWS (EC2, ECS, EKS, S3, Lambda, RDS, CloudWatch), Docker, Kubernetes, Terraform, GitHub Actions, CI/CD, Prometheus, Grafana, Linux
ML & LLM Engineering: PyTorch, CUDA, Hugging Face Transformers, RAG, FAISS, LangGraph, MCP, LLM Inference

Work Experience

Tata Consultancy Services (TCS) Apr 2021 – Oct 2023
Software Engineer | Client: GE *Remote / Hybrid*

- Developed scalable **RESTful APIs and backend microservices** using **Node.js, Express.js, and TypeScript**, implementing authentication, request validation, and database integrations.
- Led backend migration from **Node.js to Go**, addressing high-concurrency bottlenecks in a real-time monitoring system and reducing memory usage by **40%** while increasing concurrent connection capacity by **3x**.
- Built concurrent **Go services** using goroutines, channels, and worker pools to ingest and fan out **real-time financial data streams**, maintaining low-latency delivery to financial dashboards under peak load.
- Reduced **p95 API latency by 35%** by introducing a **Redis caching layer**, optimizing SQL queries and indexes, and eliminating N+1 query patterns across high-traffic endpoints.
- Designed **fault-tolerant background processing** with retries, idempotent handlers, and dead-letter queues, ensuring reliable processing of **millions of records** without data loss or duplication.
- Improved production reliability through **unit and integration testing**, **CI/CD automation**, Docker containerization, production monitoring, incident response, and root-cause analysis.
- Owned services end to end: participated in **design and code reviews**, authored **OpenAPI documentation**, and resolved **production incidents** through on-call rotations and root-cause analysis.

Launchpad Technology May 2020 – Apr 2021
Backend Software Engineer *Mumbai, India*

- Built and maintained **10+ API endpoints** for data management, user authentication, and business operations, implementing request validation, error handling, and reusable service components.
- Deployed and maintained backend applications on **AWS**, working with **EC2, S3, Linux servers, and cloud infrastructure** to support reliable application hosting and deployment.
- Designed **relational database schemas** and optimized queries with **indexing and pagination**, cutting response times on data-heavy endpoints and supporting growth in users and data volume.
- Wrote **unit and integration tests** for API endpoints and participated in **code reviews** using Git-based workflows, catching regressions early and keeping sprint releases stable.

Education

California State University, Chico | *M.S. in Computer Science* Chico, CA | May 2026
University of Mumbai | *B.E. in Computer Engineering* Mumbai, India | May 2020

Projects

CoderAI – Open-Source Multi-Agent Code Review & Repair CLI *Python, MCP, Multi-Agent Systems, Docker*

- Built a **Planner/Coder/Reviewer agent CLI** that localizes bugs and patches code, using **MCP** for tool calls and sandboxed tests.
- Ranked **5th of 30** on the Martian Code Review Benchmark (50 real PRs): **58.2% precision, 50.8% F1**, 1.1 false positives/PR, 6.6s/review.

AI Context Router | *Winner, Wefunder* *FastAPI, Python, Docker, LLMs*

- Built a routing engine that dispatches prompts across **multiple LLM providers** based on reasoning complexity, latency target, and token cost, **cutting inference cost by 15%** in testing.
- Developed async **FastAPI** services in **Docker** with health checks and **automatic failover** when a provider errors or times out.

GPUShare: Multi-Tenant A100 Scheduler for Student Workloads *Python, Docker, NVIDIA MIG*

- Built a **Python-based GPU scheduler** that partitions a shared **A100 80GB GPU** into up to **7 isolated MIG instances**, enabling concurrent execution of containerized student workloads.
- Implemented **fair-share scheduling and per-user GPU quotas** for **30 students**, preventing resource monopolization, reducing CUDA out-of-memory failures, and improving GPU resource utilization.